

Programming Homework: Decision Trees with Scikit-learn

Problem 1: Classification with Decision Trees using ID3 and C4.5

Requirements:

Using the `PlayTennis` dataset (available in the slides) or the `Iris` dataset from scikit-learn, perform the following tasks:

- a) Data preprocessing: encode categorical features (if any) into numerical values.
- b) Train a Decision Tree Classifier model using:
 - The `entropy` criterion (ID3)
 - The `log_loss` criterion (equivalent to entropy in scikit-learn)
- c) Compare the results with a model using `gini` (CART).
- d) Visualize the decision tree and explain the meaning of the nodes.
- e) Evaluate the model using cross-validation and report the accuracy.

Hints:

- Use `DecisionTreeClassifier` from `sklearn.tree`.
- Use `plot_tree` to plot the tree.
- Use `cross_val_score` for evaluation.

Problem 2: Decision Tree Regression and Handling Missing Values

Requirements:

Using the Boston Housing or Diabetes dataset from scikit-learn, perform the following tasks:

- a) Train a Decision Tree Regressor model using the `squared_error` criterion.
- b) Experiment with different tree depths and evaluate the performance using MAE and RMSE.
- c) Simulate missing data by randomly dropping 10% of the values in a feature.
- d) Compare two methods for handling missing values:
 - Imputing with the mean value
 - Using `SimpleImputer` from scikit-learn
- e) Evaluate the impact of missing data handling on model quality.

Hints:

- Use `DecisionTreeRegressor`.
- Use `SimpleImputer` or `KNNImputer` to handle missing values.
- Evaluate using `mean_absolute_error` and `mean_squared_error`.

References

- Scikit-learn Documentation: <https://scikit-learn.org/stable/modules/tree.html>
- Lecture Slides: `DecisionTree.pdf`